

Bachelor's Thesis Proposal

Personal Data Analysis of Website Privacy Notices

Assessor: Dr. Hoang Nguyen
Co-assessor: To be decided

1 Motivation

Privacy policies are the main source of information about how a website handles personal data: which data it collects, for what purposes, with whom it shares the data, and how long the data is retained. Data protection laws, such as the GDPR [2] and the Vietnamese Law on Personal Data Protection [3], require organizations to provide this information transparently.

In practice, however, privacy policies are long, written in legal language, and often vague (e.g., “we may share your information with our partners”). Very few users actually read them. Moreover, a user may have accounts on dozens of websites yet have no overview of which organizations hold their personal data, e.g., their phone number or location, and to which third parties these data are passed on.

Research tools, such as PoliGraph [1], already extract data collection statements from privacy policies and represent them as a knowledge graph. These tools can be used by website users who want to explore and compare the policies of the websites they use.

2 Objective

The objective of this thesis is to develop a web application that analyzes privacy policies and organizes their content in the form of a *graph*.

Users can also correct or complete the information in the graph manually. Since privacy policies change over time, the application should keep the graph up to date: users can trigger a rescan when they know that a policy has changed, and the application periodically rescans the stored policies on its own.

3 Tasks

1. Review the existing approaches to automated privacy policy analysis.
2. Design a graph model that captures the information extracted from privacy policies. Ideally, this model should capture the categories of personal data collected, the purposes of processing, the retention period, and the storage location.

3. Implement the privacy policy analysis so that it takes a privacy policy notice and returns a subgraph of the given model.
4. Implement a web application in which users can add websites and their privacy policies, browse, and query the graph.
5. Validate the extraction, browsing, and querying functionality by applying the system to a selection of open-source websites.
6. Document the design, implementation, and evaluation in the final report.

4 Deliverables

The following deliverables are due at the end of the project:

Final report Please have a look at the submission requirements in the following section. The final report should consist of an introduction, a (theoretical) background section, sections describing the work packages and their outcomes, and a conclusion. The report must be written in English.

Source code Complete, documented source code with a reproducible evaluation.

Presentation At the end of the thesis, a 20-minute presentation must be given in a thesis defense seminar. It should give an overview and discuss the most important highlights of the work.

References

- [1] Hao Cui, Rahmadi Trimananda, Athina Markopoulou, and Scott Jordan. PoliGraph: Automated privacy policy analysis using knowledge graphs. In *32nd USENIX Security Symposium (USENIX Security 23)*, 2023. <https://www.usenix.org/conference/usenixsecurity23/presentation/cui>.
- [2] European Parliament and Council. Regulation (EU) 2016/679, 2016. <https://eur-lex.europa.eu/eli/reg/2016/679/oj>.
- [3] National Assembly of the Socialist Republic of Vietnam. Law on personal data protection, law no. 91/2025/QH15, 2025. <https://english.luatvietnam.vn/dan-su/law-on-personal-data-protection-law-no-91-2025-qh15-405135-d1.html>.